8 Sadly, this line of investigation did not end well for the pigeons, who were in fact innocent of all wrongdoing.

9 Without knowing anything about this story except the bit about the pigeons, I happened to run into Bernard Burke a few years ago at MIT. We were just chatting as physicists tend to do, and he was telling me about some past work I didn't really follow and at some point I realized he was talking about his phone call with THE Penzias and just casually dropping the fact that he was a catalyst for one of the most important discoveries in the history of physics. A similar thing happened to me at a conference several years ago when I met Tom Kibble, who built much of the theory around the Higgs boson. Moral of the story: listen to the emeritus profs; they might have quietly revolutionized your whole field of study.

10 In the course of writing this book, I was thrilled to hear that Peebles won a 2019 Nobel Prize for, in part, the theory side of this discovery. So maybe there is some justice in the end. Just not for the pigeons.

11 The name "blackbody" comes from the idea of an object—a "body"—that perfectly absorbs all the light that hits it and reemits it as pure heat. Most objects don't do this perfectly, of course; they reflect a bit of the light, and some is absorbed without being reemitted. But most materials when heated up will glow at some level in a way that is recognizable as an approximate shape of a blackbody curve.

12 The light is all in the microwave part of the spectrum, so "redder" means lower-frequency microwave radiation, and "bluer" means higher-frequency microwave radiation, but when we make maps we do actually use colors like red and blue, because, you know, human eyes.

13 Named for Max Planck, an early originator of quantum theory. There's also a Planck energy, length, and mass, all defined through various combinations of fundamental constants, one of which is the Planck constant, which is central to anything with a